Validating and Constraining Neurovascular Coupling with Differentiable Biophysical Modeling from Tissue Microstructure to Cortical Dynamics in the WAND Multi-Modal Dataset

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Abstract

Whole-brain computational models of neural dynamics require structural connectivity, conduction delays, metabolic context, and electrophysiological recordings as independent constraints. Existing studies typically lack one or more of these measurements, forcing free-parameter assumptions that limit predictive validity. The Welsh Advanced Neuroimaging Database (WAND) [Mayerand et al., 2025] provides all of them in 170 healthy volunteers: ultra-strong-gradient diffusion MRI (300 mT/m AxCaliber), quantitative MRI (T1/T2/QMT/MP2RAGE), perfusion (pCASL, TRUST), spectroscopy (GABA/glutamate), 275-channel MEG, and TMS-EEG.

We introduce **NeuroJAX**, a differentiable, GPU-accelerated Python toolkit built on JAX [Bradbury et al., 2018] and Equinox [Kidger and Garcia, 2021] that unifies preprocessing and biophysical modelling in a single computational graph. NeuroJAX enables end-to-end gradient descent from sensor-level error through hemodynamic forward models to per-region biophysical parameters.

This paper presents the quantitative structural processing pipeline for the reference WAND subject (sub-08033) and the first set of derived maps, demonstrating that every free parameter in a standard neurovascular coupling model can be independently measured or tightly constrained.

1 Introduction

Large-scale computational models of brain dynamics [Deco et al., 2013, Riera et al., 2006] depend on three categories of subject-specific input: (1) structural connectivity and conduction delays from diffusion MRI, (2) regional tissue properties—myelination, metabolite concentrations, perfusion—that constrain biophysical parameters, and (3) functional recordings (MEG, fMRI, TMS-EEG) that provide the fitting targets. The tension in the field is that most datasets provide only a subset: the Human Connectome Project gives excellent diffusion and fMRI but no TMS; clinical TMS-EEG studies rarely include microstructural MRI.

WAND [Mayerand et al., 2025] closes this gap with seven imaging sessions per subject on the Connectom 300 mT/m scanner at CUBRIC Cardiff, covering every modality needed for a fully-constrained neurovascular model.

1.1 Differentiable generative modelling with the JAX ecosystem

Generative models in neuroimaging. The generative modelling approach to neuroimaging inverts a forward (biophysical) model to infer latent parameters from observed data. Rather than treating brain images as static maps, generative models simulate the physical processes that produce the measurements — neural mass dynamics, hemodynamic coupling, electromagnetic

forward projection, MR signal formation — and fit the model parameters to match empirical recordings. This approach has driven two decades of progress in dynamic causal modelling (DCM), whole-brain network simulation via The Virtual Brain (TVB) [Sanz Leon et al., 2013], and more recently WhoBPyT [Griffiths et al., 2024], which recasts whole-brain simulation as a deep network in PyTorch for gradient-based fitting. The limitation of these tools is that they treat preprocessing and modelling as separate stages, and their fitting procedures (Euler integration, finite-difference gradients, grid search) do not scale to the parameter dimensionality that multi-modal constraints demand.

Simulation-based inference and neural differential equations. Two developments in machine learning transform what is computationally feasible. First, **simulation-based inference** (SBI) [Cranmer et al., 2020] replaces likelihood evaluation with learned neural density estimators trained on simulated data, enabling Bayesian posterior estimation for models with intractable likelihoods — exactly the situation in whole-brain simulation where the mapping from parameters to observables passes through stiff ODEs and nonlinear forward models. When coupled with GPU-accelerated simulators, SBI makes it practical to explore parameter spaces of hundreds of dimensions that were previously inaccessible. Second, **neural ordinary differential equations** (Neural ODEs) [Chen et al., 2018] and their extensions to controlled and stochastic differential equations (CDEs, SDEs) [Kidger, 2022] embed neural networks within differential equation solvers, enabling learned dynamics that respect physical structure while remaining fully differentiable. The parallel development of **physics-informed neural networks** (PINNs) [Raissi et al., 2019] — which enforce PDE constraints as soft penalties in the loss function — provides a meshless alternative to FEM for field problems (conductivity, elasticity, electromagnetic forward models). Together, these methods enable a new class of brain model: one where the tissue-level physics, the network dynamics, and the observation forward models are all differentiable, composable, and GPU-accelerated.

The JAX + Equinox + Diffrax stack. NeuroJAX is built on JAX [Bradbury et al., 2018] with Equinox [Kidger and Garcia, 2021] for neural network modules and Diffrax for differential equation solvers — the same stack developed by Kidger for Neural CDEs. This choice is deliberate: JAX provides `jax.grad` (automatic differentiation through arbitrary Python), `jax.vmap` (automatic vectorisation for batch processing), `jax.jit` (XLA compilation to GPU/TPU), and `jax.lax.scan` (efficient sequential computation). Equinox provides PyTree-based neural network modules that compose naturally with JAX transformations. Diffrax provides adaptive-step ODE/SDE/CDE solvers with adjoint backpropagation. The result is that the entire forward model — from Bloch equation tissue simulation through Balloon–Windkessel hemodynamics to sensor-level prediction — compiles into a single XLA computation graph through which gradients flow end-to-end:

$$\mathcal{L}(\theta) = \sum_m w_m \, d_m(\mathbf{y}_m^{\text{sim}}(\theta;\, \mathbf{C}, \mathbf{D}, \mathbf{L}),\ \mathbf{y}_m^{\text{emp}}) \tag{1}$$

where θ are biophysical parameters, $\mathbf{C}/\mathbf{D}$ are the structural connectome weights/delays, $\mathbf{L}$ is the leadfield, and m indexes modalities (FC, FCD, TEP waveform, sensor MSE). This enables gradient-based fitting with per-region heterogeneous parameters across hundreds of brain regions — a problem that grid search or finite-difference methods cannot touch, but that `jax.grad` solves in seconds on a GPU.

2 Dataset

2.1 CUBRIC and the Connectom scanner

The Cardiff University Brain Research Imaging Centre (CUBRIC) has been at the forefront of human connectomics since the installation of a Siemens MAGNETOM Connectom 3T scanner

with 300 mT/m gradients in 2016 — the first in Europe and only the second worldwide after the Massachusetts General Hospital system built for the Human Connectome Project [Jones et al., 2022]. Under the direction of Derek K. Jones, CUBRIC has used these ultra-strong gradients to map white matter microstructure at unprecedented precision, enabling AxCaliber axon diameter estimation and CHARMED restricted-fraction imaging in vivo.

Complementing the diffusion programme, Krish D. Singh's group has pioneered the combination of magnetic resonance spectroscopy with electrophysiology [Muthukumaraswamy et al., 2009], demonstrating that resting GABA concentration in visual cortex predicts peak gamma oscillation frequency and fMRI BOLD amplitude. This established a direct, measurable link between inhibitory neurotransmitter concentration and neural dynamics — a founding principle of the WAND dataset design and the MRS-constrained neural mass modelling in this work. The Singh group's subsequent studies extended this to glutamate, schizophrenia risk markers, and motor cortex, building the empirical case that MRS metabolites can serve as independent constraints on computational models of cortical circuit function.

WAND [Mayerand et al., 2025] synthesises these programmes into a single cohort of 170 healthy volunteers (aged 18–63, BIDS-organised), acquired across up to 8 sessions on three scanner platforms at CUBRIC. The dataset was published in *Scientific Data* (2024) and is available via GIN (`https://gin.g-node.org/CUBRIC/WAND`). To our knowledge, no computational modelling studies have yet been published using WAND; the present work represents the first application of differentiable biophysical modelling to this resource.

2.2 Scanner platforms and sessions

The WAND sessions span four distinct instrument platforms:

- **Siemens MAGNETOM Connectom 3T** (300 mT/m, ses-02, ses-06): Ultra-strong gradients for diffusion microstructure (AxCaliber, CHARMED) and quantitative relaxometry (VFA, QMT). Sub-mm structural imaging (MP2RAGE 0.7 mm, 7-echo GRE 0.67 mm).
- **Siemens MAGNETOM Prisma 3T** (80 mT/m, 32-channel receive-only head coil, ses-03, ses-05): fMRI, perfusion (pCASL, TRUST), structural (T1w, T2w), and MRS (semi-LASER).
- **Siemens MAGNETOM 7T** (32-channel receive / volume transmit head coil, Nova Medical, ses-04): High-field MRS (semi-LASER) for improved spectral resolution of GABA and glutamate.
- **CTF 275-channel radial gradiometer MEG** (ses-01): Sampled at 1200 Hz, analysed as synthetic third-order gradiometers with 29 reference channels. EOG recorded concurrently. Five paradigms: resting-state, auditory-motor, mismatch negativity, Simon task, visual.
- **Magstim BiStim^2 TMS** (ses-08, Magstim, Whitland, UK): D70 Alpha Flat figure-of-eight coil (70 mm). Neuronavigated with BrainSight (Rogue Research, Montreal) and Polaris optical tracking (Northern Digital). Controlled via CED 1401 interface with Signal v6.04a software. EMG recorded with Digitimer D440-4 amplifier and pre-gelled Ag/AgCl electrodes. SICI paired-pulse protocol, 3 runs per subject.

Table 1 details the acquisitions per session for the reference subject sub-08033 (all modalities confirmed, 10.24 GB).

2.3 CUBRIC-provided derivatives

In addition to raw BIDS data, CUBRIC distributes several pre-computed derivatives:

Table 1: WAND acquisition sessions for sub-08033.

Session	Scanner	Modalities	Key acquisitions
ses-01	CTF MEG	MEG	275ch: resting, auditory-motor, MMN, Simon, visual + headshape
ses-02	Connectom 3T	DWI + anat	AxCaliber (4 shells, b=0–15500), CHARMED (AP+PA), T1w, VFA (SPGR/SSFP/IR), QMT (16 vols)
ses-03	Prisma 3T	fMRI + perf	T1w, T2w (grad-corrected), fieldmaps, pCASL, TRUST, InvRec, angiography, M0
ses-04	7T	MRS + anat	semi-LASER: ACC, occipital, auditory, sensorimotor; T1w
ses-05	Prisma 3T	MRS + anat	semi-LASER (test–retest); MEGA-PRESS; T1w
ses-06	Connectom 3T	Structural	MP2RAGE (0.7 mm), 7-echo GRE (0.67 mm mag+phase), T2w (0.21 mm in-plane)
ses-08	TMS lab	TMS-EEG	SICI protocol (3 runs, .mat format)

- **Brain-extracted T1w images** via HD-BET for each session
- **Diffusion QC**: FSL `eddy_qc` reports (motion parameters, outlier counts, CNR maps) for the multishell DWI
- **Structural and functional QC**: MRIQC reports for T1w, T2w, and BOLD data
- **MEG–MRI coregistrations**: T1w images transformed to CTF MEG coordinate space (Cardiff GitLab)
- **Phenotype data**: age, sex, ethnicity, anthropometrics (weight, height, waist/hip, blood pressure), alcohol and smoking status (`participants.tsv`)
- **Cognitive and behavioural data**: trait questionnaires and task performance (`phenotype/` directory)

The processing reported here extends these CUBRIC derivatives in three directions: (1) quantitative tissue property maps from the rich relaxometry acquisitions (VFA T1, QMT BPF, T2*, g-ratio, MP2RAGE T1); (2) advanced subcortical segmentations (DTI-enhanced thalamic nuclei, hippocampal subfields, hypothalamic subunits); and (3) perfusion quantification (CBF, OEF, $CMRO_2$). We also extend the diffusion and microstructural derivatives beyond CUBRIC's eddy QC reports with DTI fitting (FA, MD, V1 from the CHARMED low-b shells), and lay the groundwork for AxCaliber axon diameter estimation and microstructure-informed connectome construction via sbi4dwi on the full ultra-strong-gradient data.

3 Methods

The breadth of the WAND dataset demands a correspondingly broad processing toolkit. Two complementary open-source projects — FreeSurfer and FSL — form the backbone of our pipeline, together covering every modality in the dataset.

FreeSurfer [Fischl, 2012] grew out of the cortical surface reconstruction work of Anders Dale and Martin Sereno at UC San Diego in the early 1990s [Dale et al., 1999, Fischl et al., 1999].

Bruce Fischl, joining as a postdoctoral fellow, replaced the manual steps of the surface pipeline with automated algorithms; the collaboration continued as Fischl and Douglas Greve built FreeSurfer into a production tool at the Athinoula A. Martinos Center for Biomedical Imaging, Massachusetts General Hospital, where it has been developed and maintained ever since. Over two decades it has grown into the standard tool for cortical parcellation, subcortical segmentation, and surface-based analysis. Recent versions (8.x) add deep-learning modules — SynthSeg for contrast-invariant segmentation [Billot et al., 2023], SynthMorph for contrast-agnostic registration [Hoffmann et al., 2022], and CNN-based thalamic and hippocampal segmentation — that extend its reach to the heterogeneous multi-contrast data in WAND.

FSL (FMRIB Software Library) [Jenkinson et al., 2012, Smith et al., 2004] was first released in 2000 from the Oxford Centre for Functional MRI of the Brain (FMRIB) and has become the most widely used neuroimaging analysis suite, with tools spanning structural (FAST, FIRST, fsl_anat), diffusion (eddy, bedpostx, probtrackx2, xtract, dtifit), functional (FEAT, MELODIC, FIX), and registration (FLIRT, FNIRT, MMORF). Critically for WAND, FSL has extended beyond its MRI core into two additional domains: **perfusion imaging**, via oxford_asl and BASIL/FABBER [Chappell et al., 2009] for ASL quantification, and **magnetic resonance spectroscopy**, via FSL-MRS [Clarke et al., 2021] for end-to-end SVS and MRSI analysis. Through the Oxford Centre for Human Brain Activity (OHBA), the FSL ecosystem further extends to **electrophysiology**: osl-ephys provides MEG/EEG preprocessing and source reconstruction [van Es et al., 2025], while osl-dynamics implements HMM and DyNeMo brain state models [Gohil et al., 2023]. This remarkable convergence means that a single software ecosystem — FSL + OHBA — can process structural MRI, diffusion, fMRI, perfusion, spectroscopy, and MEG from WAND, with FreeSurfer providing the complementary cortical surface infrastructure and a host of tools for advanced structural processing.

NeuroJAX and vpjax add the differentiable modelling layer that these preprocessing tools lack: end-to-end gradient-based fitting from tissue-level biophysics through hemodynamics to sensor-level predictions, all compiled in JAX for GPU acceleration.

3.1 FreeSurfer recon-all + advanced segmentations

Standard cortical reconstruction was performed with FreeSurfer 8.2.0 (`recon-all` with T2-weighted pial refinement from ses-03). Additional segmentations:

- **SAMSEG** [Puonti et al., 2016]: joint T1w+T2w Bayesian segmentation, providing tissue probability maps and intracranial volume.
- **SynthSeg** [Billot et al., 2023]: contrast-invariant DL segmentation applied to T1w from ses-02, 03, 04, 05 for cross-session volumetric consistency.
- **Hypothalamic subunits**: DL segmentation (Billot et al. 2020).
- **Thalamic nuclei**: FreeSurfer atlas-based segmentation with optional DTI enhancement.
- **Hippocampal subfields + amygdala nuclei**: with T2w for improved subfield boundaries.

3.1.1 SAMSEG tissue volumes

3.1.2 Cross-session volumetrics (SynthSeg)

3.1.3 Hypothalamic subunits

3.2 T1w/T2w myelin ratio

The T1w/T2w ratio [Glasser and Van Essen, 2011] was computed after rigid-body registration of the gradient-corrected T2w to T1w (FLIRT 6-DOF), SynthStrip skull stripping, and voxelwise

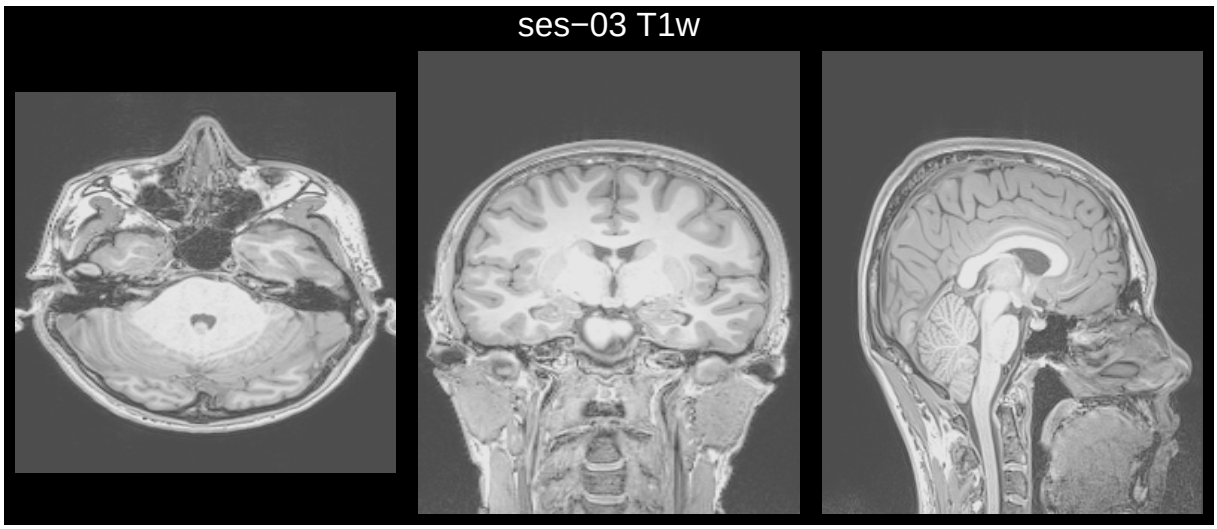

Figure 1: Structural T1w (ses-03) for sub-08033. The T1-weighted image provides the anatomical reference for all cross-session registrations and serves as input for FreeSurfer cortical reconstruction, tissue segmentation (FAST/SAMSEG), and BEM forward model generation for MEG/EEG source localisation.

Table 2: SAMSEG tissue volumes (T1w+T2w joint segmentation).

Structure	Volume (mm^3)
Soft_Nonbrain_Tissue	1,394,952
CSF	374,654
Left-Cerebral-Cortex	268,149
Right-Cerebral-Cortex	266,766
Right-Cerebral-White-Matter	246,502
Left-Cerebral-White-Matter	239,711
Left-Cerebellum-Cortex	59,600
Right-Cerebellum-Cortex	59,053
Skull	49,736
Brain-Stem	22,098
Left-Cerebellum-White-Matter	14,864
Right-Cerebellum-White-Matter	14,666

Table 3: Cross-session volumetrics (SynthSeg, mm^3).

Structure	ses-02	ses-03	ses-04	ses-05
left cerebral white matter	280,441	280,766	275,760	282,130
right cerebral white matter	277,799	280,164	271,761	280,626
left cerebral cortex	321,868	322,111	319,239	324,713
right cerebral cortex	317,842	324,973	322,051	324,671
left hippocampus	4,493	4,646	4,336	4,654
right hippocampus	4,045	4,278	4,062	4,131
left thalamus	9,307	9,244	9,404	9,408
right thalamus	9,447	9,649	9,675	9,723

Table 4: Hypothalamic subunit volumes (mm^3).

Subunit	Volume (mm^3)
left anterior-inferior	18.8
left anterior-superior	29.1
left posterior	152.2
left tubular inferior	137.6
left tubular superior	124.4
right anterior-inferior	21.9
right anterior-superior	30.1
right posterior	156
right tubular inferior	131.1
right tubular superior	143.9
whole left	462.2
whole right	483.1

division. The ratio was projected to the cortical mid-surface (`mri_vol2surf`, 0.2–0.8 cortical depth).

3.3 VFA T1 mapping (DESPOT1)

Quantitative T1 was estimated from the ses-02 SPGR variable flip angle data (8 flip angles: 2–18°, $\mathrm{TR}_{\mathrm{exc}} = 4\,\mathrm{ms}$) using the linearised DESPOT1 method [Deoni et al., 2003]:

$$\frac{S}{\sin\alpha} = E_1 \frac{S}{\tan\alpha} + M_0(1 - E_1), \qquad E_1 = e^{-\mathrm{TR}/T_1} \tag{2}$$

3.4 QMT bound pool fraction

Quantitative magnetization transfer was fitted from 11 MT-on volumes (3 MT flip angles × multiple offset frequencies: 1–56 kHz) plus one MT-off reference, using a two-pool Ramani model [Ramani et al., 2002] with super-Lorentzian lineshape for the bound pool. The bound pool fraction (BPF) provides a direct measure of macromolecular (myelin) content [Sled and Pike, 2001].

3.5 Multi-echo GRE: T2* and R2* mapping (ses-06)

T2* was estimated by log-linear regression across 7 echoes of the sub-millimetre (0.67 mm) multi-echo GRE from ses-06. R2* = 1/T2* is sensitive to both iron and myelin content; combined with QSM (from the phase data, not yet processed), iron and myelin contributions can be separated.

3.6 MP2RAGE quantitative T1 (ses-06)

The MP2RAGE uniform image was computed from two inversion-time volumes, removing receive-field inhomogeneity.

3.7 Quantitative perfusion processing

The full perfusion pipeline combines three independent measurements — cerebral blood flow, venous oxygenation, and blood T1 — to yield the cerebral metabolic rate of oxygen consumption ($\mathrm{CMRO_2}$). CBF is quantified with FSL's `oxford_asl`, which wraps the BASIL (Bayesian Inference for Arterial Spin Labelling) framework [Alsop et al., 2015]. BASIL itself uses FABBER,

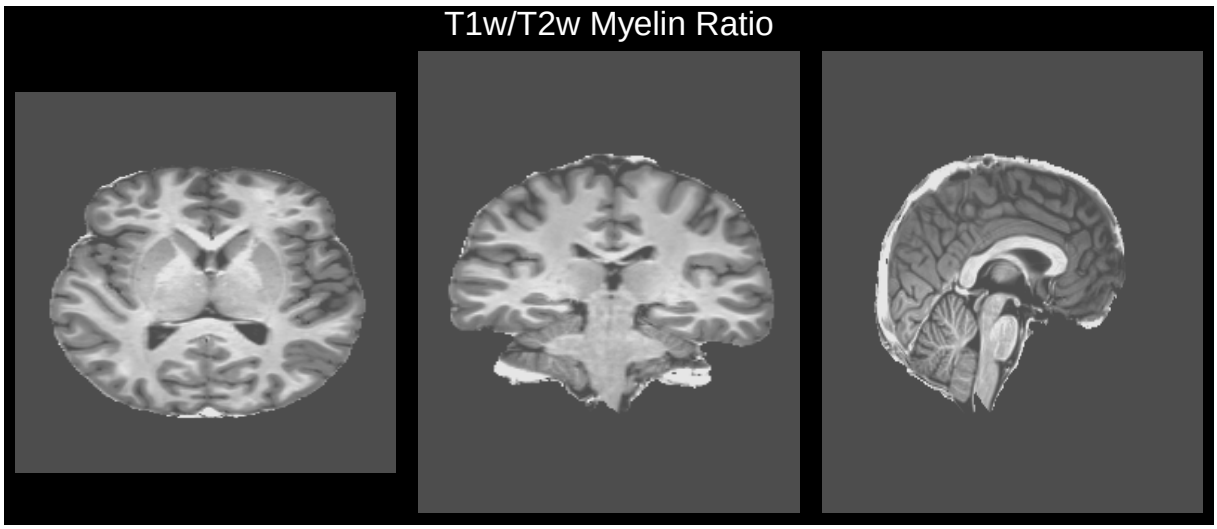

Figure 2: T1w/T2w myelin ratio (ses-03). This ratio serves as a widely-used proxy for cortical myelination, indexing the hierarchical organisation of cortex from heavily myelinated primary sensory areas to lightly myelinated association cortex. WAND uniquely allows validation of this proxy against ground-truth QMT bound pool fraction (Fig. 4) and multi-echo GRE R2* (Fig. 5).

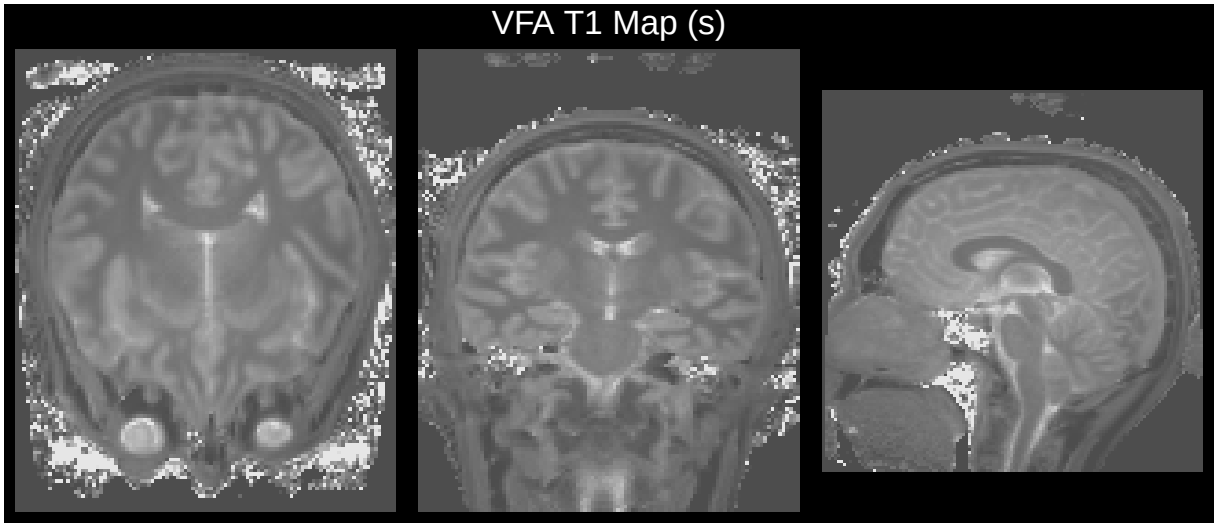

Figure 3: Quantitative T1 map from ses-02 SPGR variable flip angle data (DESPOT1). Quantitative T1 reflects tissue water content and macromolecular environment: white matter (short T1, dark) is distinguished from grey matter (longer T1, bright). This map serves as a required input for the QMT two-pool model and provides an independent measure of tissue composition alongside the T1w/T2w ratio.

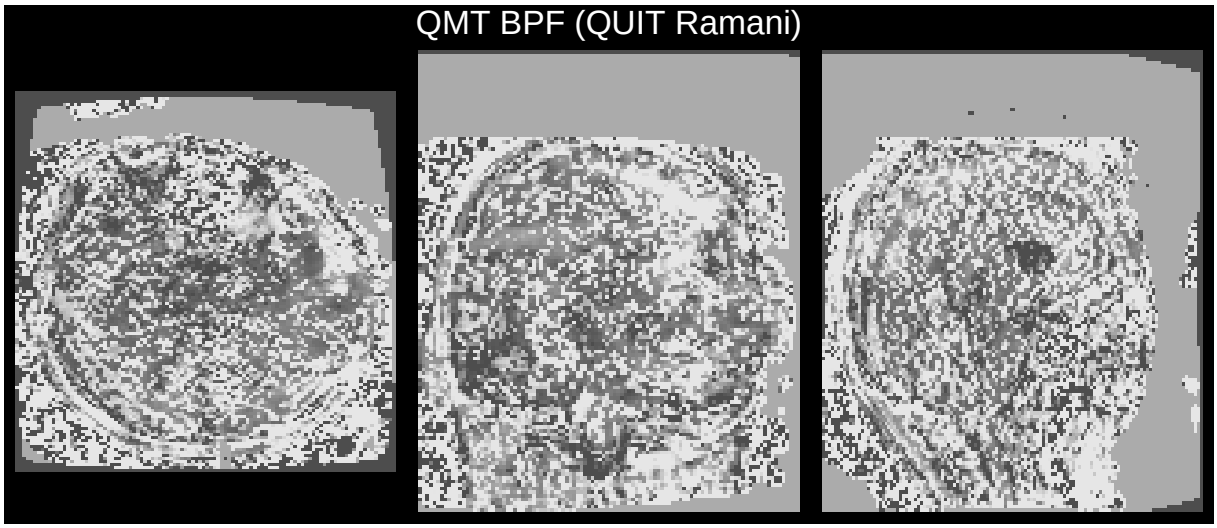

Figure 4: QMT bound pool fraction (ses-02, QUIT Ramani model with super-Lorentzian lineshape). BPF directly measures macromolecular (myelin) content via the restricted proton pool, providing the ground-truth measurement against which the T1w/T2w myelin proxy is validated. Median BPF = 0.10, consistent with healthy white matter. Combined with AxCaliber axon diameter, BPF yields the g-ratio and thence per-tract conduction velocity via the Hursh–Rushton relationship.

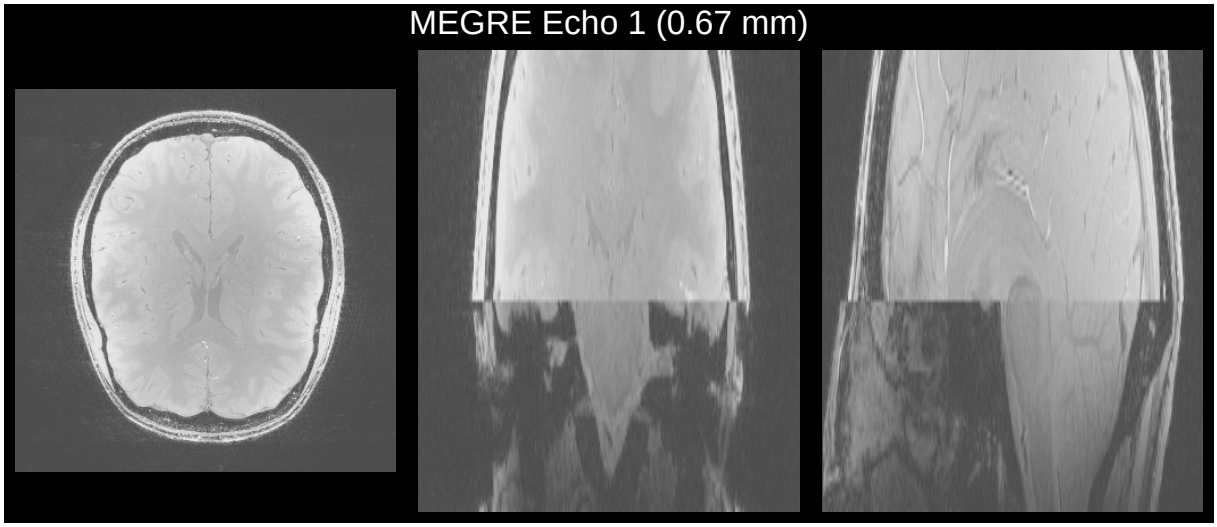

Figure 5: Multi-echo GRE first echo magnitude (ses-06, 0.67 mm isotropic). The sub-millimetre resolution enables cortical depth-resolved analysis via LAYNII or Nighres. The 7-echo magnitude data yields R2* maps sensitive to iron and myelin; the phase data (not yet processed) enables QSM for separating paramagnetic iron from diamagnetic myelin contributions — critical for resolving the feedforward/feedback cortical hierarchy.

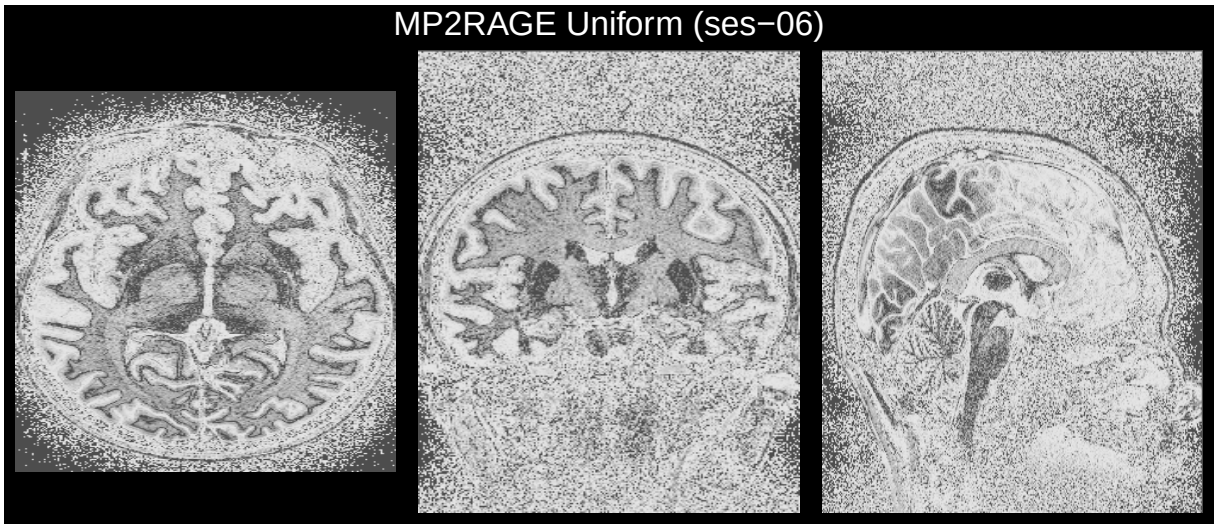

Figure 6: MP2RAGE uniform image (ses-06, 0.7 mm isotropic). The MP2RAGE sequence provides B1-corrected quantitative T1 at sub-millimetre resolution, complementing the ses-02 VFA T1 map. Cross-validation between two independent T1 measurements (VFA and MP2RAGE) from different sessions and scanners tests the stability of quantitative tissue property estimation.

FSL's variational Bayesian inference engine, to invert the kinetic model with spatial regularisation and partial-volume correction. OEF is obtained from TRUST, and subject-specific blood T1 from the Look-Locker inversion recovery acquisition. These combine via Fick's principle to give $CMRO_2$.

3.7.1 CBF from pCASL

Cerebral blood flow was quantified from pseudo-continuous ASL (bolus = 1.8 s, PLD = 2.0 s, 110 volumes) using FSL `oxford_asl` [Alsop et al., 2015] with voxelwise M0 calibration, motion correction, and partial-volume correction. The ASL white paper single-compartment model [Alsop et al., 2015] gives:

$$\mathrm{CBF} = \frac{6000 \cdot \lambda \cdot (\overline{S}_{\mathrm{ctrl}} - \overline{S}_{\mathrm{label}}) \cdot e^{\mathrm{PLD}/T_{1b}}}{2 \cdot \alpha \cdot T_{1b} \cdot M_0 \cdot (1 - e^{-\tau/T_{1b}})} \quad [\mathrm{ml/100g/min}] \tag{3}$$

where $\lambda = 0.9\,\mathrm{ml/g}$ is the blood–brain partition coefficient, $\alpha = 0.85$ the labelling efficiency, $\tau = 1.8\,\mathrm{s}$ the bolus duration, and T_{1b} the subject-specific blood T1.

Blood T1 from inversion recovery. Subject-specific T_{1b} was estimated from the Look-Locker acquisition (128×128×1, 960 time points, TR = 150 ms). The signal in a sagittal sinus ROI was fitted to:

$$S(t) = A\big(1 - B\, e^{-t/T_1^*}\big), \qquad T_1 = T_1^* \cdot B \tag{4}$$

where the Deichmann–Haase correction recovers true T_1 from the apparent T_1^* and the inversion efficiency B.

3.7.2 TRUST: venous oxygenation

T2 of venous blood in the sagittal sinus was estimated from the TRUST acquisition [Lu and Ge, 2008] (24 volumes, 4 effective TEs × 3 repeats × control/label). The control–label difference signal in the sinus decays as:

$$|\Delta S|(\mathrm{eTE}) = S_0\, e^{-\mathrm{eTE}/T_2} \tag{5}$$

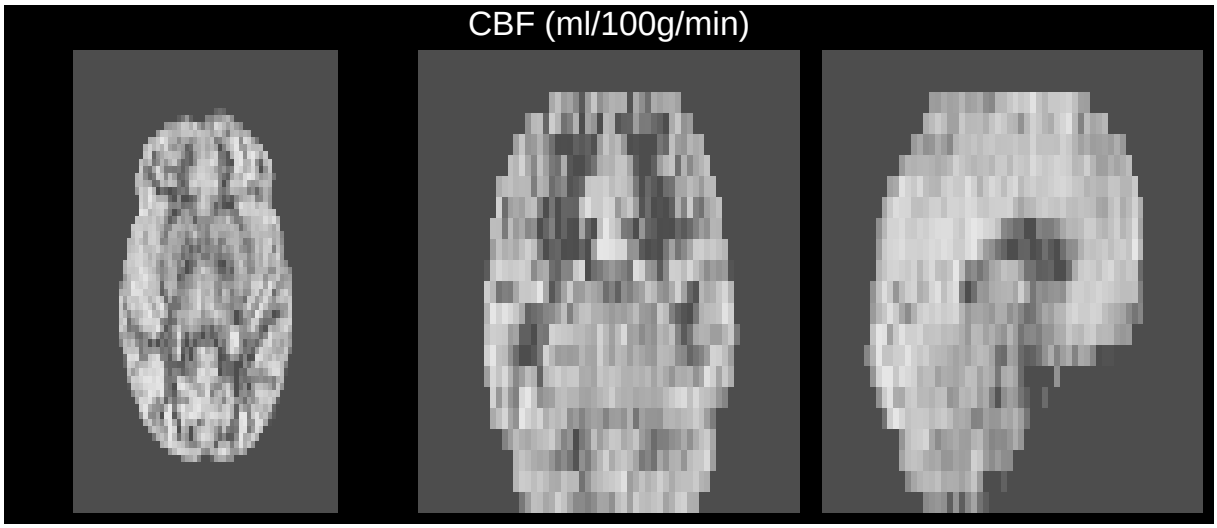

Figure 7: Calibrated CBF map from pCASL (ses-03, ml/100g/min). Cerebral blood flow is the first of three independent perfusion measurements in WAND. Combined with OEF from TRUST and CaO_2 from haematocrit, CBF enters Fick's equation to yield $CMRO_2$ — the metabolic demand signal that drives the Balloon–Windkessel hemodynamic model. GM mean = 113, WM mean = 24 ml/100g/min.

Venous T2 is calibrated to oxygen saturation Y via the quadratic model [Lu et al., 2012] at 3 T:

$$\frac{1}{T_2} = A + B(1-Y) + C(1-Y)^2 \tag{6}$$

with coefficients $A = 4.5$, $B = 47.1$, $C = 55.5\,\mathrm{s}^{-1}$ for Hct = 0.42. Solving the quadratic for Y gives SvO_2, from which OEF follows.

3.7.3 $CMRO_2$ via Fick's principle

The three independent perfusion measurements combine via the Fick equation to give the cerebral metabolic rate of oxygen consumption:

$$\mathrm{CMRO_2} = \mathrm{CBF} \times \mathrm{OEF} \times C_aO_2 \tag{7}$$

where $\mathrm{OEF} = (\mathrm{SaO_2} - \mathrm{SvO_2})/\mathrm{SaO_2}$ from TRUST and the arterial oxygen content is:

$$C_aO_2 = [\mathrm{Hb}] \times 1.34 \times \mathrm{SaO_2} \approx 8.8\ \mu\mathrm{mol\,O_2/ml\,blood} \tag{8}$$

for Hb = 15 g/dl and $SaO_2 = 0.98$. This hierarchy from global OEF (TRUST) can be upgraded to regional OEF via quantitative BOLD [He and Yablonskiy, 2007] fitting of the 7-echo GRE (ses-06).

3.8 Diffusion microstructure processing

Diffusion-weighted images from ses-02 (AxCaliber: 4 shells, $b = 0$–$15\,500\,\mathrm{s/mm^2}$; CHARMED: AP+PA) were preprocessed with FSL's standard distortion and motion correction pipeline. `topup` estimated the susceptibility-induced off-resonance field from the AP/PA b=0 image pair, and `eddy` corrected for eddy-current distortions, subject motion, and signal dropout (outlier replacement enabled). CUBRIC-provided `eddy_qc` reports confirmed low motion and few outlier slices for sub-08033.

DTI fitting was performed with `dtifit` on the low-b shells ($b \leq 2000\,\mathrm{s/mm^2}$), yielding fractional anisotropy (FA), mean diffusivity (MD), and the principal eigenvector (V1). Multi-fibre estimation used `bedpostx` (Bayesian Estimation of Diffusion Parameters Obtained using Sampling Techniques), modelling up to 3 crossing fibres per voxel with the `zeppelin` model for the extra-axonal compartment.

Tract segmentation employed `xtract`, FSL's automated tractography tool, which segments 42 major white-matter tracts using standardised seed/target/exclusion ROIs in MNI space, warped to native diffusion space via nonlinear registration (`fnirt`). Probabilistic tractography was run with `probtrackx2` using the bedpostx fibre orientations with 5000 streamlines per seed voxel.

Whole-brain structural connectomes ($\mathbf{C}$, $\mathbf{L}$) were constructed by seeding streamlines from each parcel of the Desikan–Killiany cortical atlas (from FreeSurfer) and counting streamline endpoints in every other parcel. The connection weight matrix $\mathbf{C}$ stores streamline counts (log-normalised) and the fibre length matrix $\mathbf{L}$ stores median streamline lengths. Combined with g-ratio from QMT BPF and the Hursh–Rushton relationship, $\mathbf{L}$ yields the conduction delay matrix $\mathbf{D}_{ij} = L_{ij}/v_{ij}$ with per-connection velocity v_{ij}.

For microstructural modelling beyond DTI, `sbi4dwi` (simulation-based inference for diffusion MRI) will fit the AxCaliber model to the full ultra-strong-gradient data, estimating axon diameter distributions and restricted signal fractions. DIPY serves as a complementary toolkit for signal reconstruction, tractography algorithm comparison, and diffusion spectrum imaging.

3.9 Resting-state fMRI processing

WAND ses-03 includes resting-state BOLD fMRI on the Prisma 3T, providing the functional connectivity targets against which structural connectivity and simulated dynamics are compared.

Preprocessing. Resting fMRI preprocessing follows the FSL pipeline: motion correction (`mcflirt`), brain extraction (`bet`), spatial smoothing, high-pass temporal filtering, and registration to standard space via the `fsl_anat` structural output (FLIRT + FNIRT). Alternatively, fMRIPrep [Esteban et al., 2019] provides a BIDS-native preprocessing workflow with automated confound regression, fieldmap correction, and surface projection via FreeSurfer.

Independent component analysis. Resting-state networks are identified using FSL's MELODIC [Beckmann and Smith, 2004], which performs probabilistic independent component analysis (ICA) on the 4D BOLD data to decompose it into spatially independent components with associated time courses. Group-level ICA across WAND subjects will identify the canonical resting-state networks that Smith et al. [Smith et al., 2009] defined by demonstrating a one-to-one correspondence between BrainMap task-activation networks and resting-state networks derived from ICA of nearly 30,000 subjects. These canonical networks — including default mode, dorsal attention, salience, visual, sensorimotor, frontoparietal, and language — serve as the functional parcellation against which whole-brain model predictions of functional connectivity (FC) and functional connectivity dynamics (FCD) are evaluated.

Artefact removal. FSL's FIX (FMRIB's ICA-based X-noisifier) provides automated classification of ICA components into signal and noise (motion, physiological, scanner artefacts), enabling data-driven denoising without the need for explicit confound regressors. For WAND, where physiological monitoring data are partially available, FIX offers a robust alternative to nuisance regression.

Dual regression and network modelling. Subject-specific network maps are obtained via dual regression against the group ICA templates, yielding per-subject spatial maps and time

courses for each canonical network. These time courses provide the empirical FC and FCD matrices that serve as fitting targets for the NeuroJAX whole-brain model (Eq. 1). Bijsterbosch, Smith, and Beckmann [Bijsterbosch et al., 2017] provide a comprehensive treatment of these methods in their Oxford Neuroimaging Primer on resting-state fMRI functional connectivity.

3.10 MEG processing

MEG data from ses-01 (CTF 275-channel, 1200 Hz, synthetic third-order gradiometers) were processed using the `osl-ephys` pipeline. Pre-processing comprised: (i) downsampling to 250 Hz, (ii) bandpass filtering (1–45 Hz for resting-state, broadband for task), (iii) automated bad-channel detection and interpolation, and (iv) ICA artifact rejection via `osl_dynamics.preprocessing`, removing ocular (EOG-correlated) and cardiac components by temporal correlation and topographic template matching.

Source reconstruction used the RHINO (Registration of Headshapes Including Nose in OSL) coregistration procedure, which aligns the Polaris-digitised headshape to the FreeSurfer scalp surface extracted from the ses-03 T1w. A single-shell boundary element model (BEM) provided the forward model for beamforming (linearly constrained minimum variance, LCMV) and minimum-norm estimation. Source time courses were extracted at each parcel centroid of the Desikan–Killiany atlas and sign-flipped to resolve the inherent source orientation ambiguity across subjects.

For dynamic brain state analysis, parcellated source envelopes were prepared with time-delay embedding (TDE, ± 7 lags) and PCA dimensionality reduction (retaining 95% variance). This spatio-temporal feature space served as input to `osl-dynamics`, which provides both the Hidden Markov Model (HMM) of Vidaurre et al. [2018] and DyNeMo [Gohil et al., 2023], a deep generative model that estimates time-varying linear mixtures of mode covariances. State/mode spectra, fractional occupancies, lifetimes, and switching rates provide the functional summary statistics against which the NeuroJAX whole-brain simulation is validated.

3.11 MRS processing

Magnetic resonance spectroscopy data were acquired with the semi-LASER (sLASER) sequence at both 3 T (ses-05, Prisma) and 7 T (ses-04). Spectra were processed with `fsl_mrs` [Clarke et al., 2021], FSL's spectroscopy quantification tool, which performs frequency-and-phase alignment, coil combination, eddy-current correction, water reference scaling, and LCModel-style basis set fitting in the frequency domain.

Voxel localisation used `svs_segment`: the MRS voxel was registered to the ses-03 T1w and tissue fractions (GM, WM, CSF) were computed from FAST segmentation within the voxel bounding box. These fractions enable water-scaled absolute quantification of metabolite concentrations (institutional units, i.u.) and partial-volume correction for tissue-specific concentrations.

The primary metabolites of interest are GABA and glutamate (Glu). Following the approach of Muthukumaraswamy et al. [2009], regional GABA concentration constrains the inhibitory synaptic coupling $J_I \propto [\text{GABA}]$ in the Reduced Wong-Wang neural mass model [Deco et al., 2013], while glutamate constrains the excitatory NMDA coupling $J_{\text{NMDA}} \propto [\text{Glu}]$. With WAND's four voxel placements (ACC, occipital, auditory, sensorimotor), these constraints cover both default-mode and primary sensory regions, reducing parameter degeneracy in the whole-brain fit. MEGA-PRESS data from ses-05 provide an additional GABA estimate for test–retest reliability assessment.

3.12 TMS-EEG processing

TMS data from ses-08 used the short-interval intracortical inhibition (SICI) paired-pulse protocol with the Magstim BiStim^2 system. SICI pairs a subthreshold conditioning stimulus (80% RMT,

2 ms ISI) with a suprathreshold test stimulus to probe GABAergic inhibition in motor cortex. Three runs were acquired per subject.

Analysis extracts TMS-evoked potentials (TEPs) from the CED-recorded EEG/EMG data, time-locked to the TMS pulse. After artifact rejection (TMS pulse artifact blanking, muscle artifact ICA removal), the TEP waveform is epoched (-100 to $+500$ ms) and baseline-corrected. Key TEP components (N15, P30, N45, P60, N100, P180) index successive stages of cortical processing: early components (<50 ms) reflect local excitability while late components (>100 ms) reflect network propagation through long-range connections.

The SICI ratio (conditioned MEP amplitude / unconditioned MEP amplitude) provides a functional readout of GABAergic inhibition that complements the MRS GABA concentration, linking neurochemistry to cortical excitability. In the NeuroJAX framework, TEP waveforms serve as fitting targets: the whole-brain model receives a simulated TMS impulse at the stimulation site and the predicted network response is compared to the measured TEP, enabling gradient-based optimisation of connection-specific parameters.

4 Results

4.1 vpjax: Differentiable vascular physiology

The quantitative structural maps feed into **vpjax** (Virtual Physiology in JAX), the vascular/metabolic complement to vbjax (Virtual Brain in JAX). Where vbjax provides neural mass models, vpjax provides the complete physiological forward chain from neural activity to measurable haemodynamic signals:

$$\underbrace{z(t)}_{\text{neural}} \xrightarrow{\text{CMRO}_2} \underbrace{f(t)}_{\text{CBF}} \xrightarrow{\text{Balloon}} \underbrace{v(t),\ q(t)}_{\text{volume, deoxy-Hb}} \xrightarrow{T_2^*} \underbrace{y(t)}_{\text{BOLD/ASL/qBOLD}} \tag{9}$$

4.1.1 Balloon–Windkessel hemodynamics

The extended Riera [Riera et al., 2006] neurovascular model replaces the simplified Buxton hemodynamics with a full metabolic intermediate chain:

$$\dot{s} = z - \kappa\, s - \gamma\,(f - 1) \tag{10}$$

$$\dot{f} = s \tag{11}$$

$$\tau\, \dot{v} = f - v^{1/\alpha} \tag{12}$$

$$\tau\, \dot{q} = f\, \frac{E(f,\, \text{OEF}_0)}{E_0} - \frac{q}{v}\, v^{1/\alpha} \tag{13}$$

where $E(f, E_0) = 1 - (1 - E_0)^{1/f}$ is the oxygen extraction fraction under the Grubb assumption, α is the Grubb exponent, κ is the signal decay rate, γ the autoregulatory feedback, and τ the transit time.

4.1.2 qBOLD: regional OEF from multi-echo GRE

The key upgrade from global OEF (TRUST) to regional OEF uses the quantitative BOLD model [He and Yablonskiy, 2007]:

$$S(\text{TE}) = S_0 \;\cdot\; \mathcal{F}(\text{OEF, DBV, TE}) \;\cdot\; e^{-R_2 \cdot \text{TE}} \tag{14}$$

where $\mathcal{F}$ models the intravoxel frequency distribution from randomly oriented deoxygenated vessels. At short TE the signal is dominated by R_2 (tissue T2); at long TE by R_2' (reversible, from deoxy-Hb). The difference $R_2' = R_2^* - R_2 \propto \text{OEF} \times \text{DBV} \times B_0$. WAND's 7-echo GRE (TE = 5–35 ms) is ideal for fitting this model.

4.1.3 $CMRO_2$ hierarchy

vpjax provides three levels of oxygen metabolism estimation:

Table 5: $CMRO_2$ estimation hierarchy in vpjax.

Level	OEF source	Spatial resolution	WAND data
1. Global	TRUST (sagittal sinus T2)	Whole-brain average	ses-03 TRUST
2. Regional	qBOLD (7-echo GRE, ses-06)	Per-voxel	ses-06 MEGRE
3. Dynamic	Riera model (Eqs. 10–13)	Time-varying	MEG + fMRI

4.1.4 Local linearization integration

The Balloon–Windkessel equations (Eqs. 10–13) are stiff: the vascular response evolves on a $\sim$1 s timescale while neural dynamics operate at $\sim$1 ms. Standard Euler or RK4 integration requires impractically small time steps. vpjax uses the **local linearization** (LL) method (Ozaki 1992, extended by Riera), which analytically solves the linearised system at each step:

$$\mathbf{x}(t + \Delta t) = e^{\mathbf{J}(\mathbf{x}_t)\,\Delta t}\,\mathbf{x}(t) + \mathbf{J}^{-1}(e^{\mathbf{J}\,\Delta t} - \mathbf{I})\,\mathbf{f}_0 \tag{15}$$

where $\mathbf{J} = \partial\mathbf{f}/\partial\mathbf{x}$ is the Jacobian evaluated at the current state. This permits large time steps (Δt up to 10$\times$ Euler) while remaining stable, and is fully differentiable via `jax.grad` through the matrix exponential, enabling gradient-based fitting of hemodynamic parameters alongside neural mass parameters in a single optimization.

4.1.5 Model variables and their independent measurements

The central claim is that WAND provides independent measurements for every state variable and parameter in the neurovascular model:

Table 6: Model variables and their independent measurements.

Model variable	vpjax module	WAND measurement
Neural activity $z(t)$	vbjax	MEG
Metabolic demand ($CMRO_2$)	`metabolism/`	pCASL $\times$ TRUST
Blood flow $f(t)$	`hemodynamics/`	pCASL CBF
Oxygen extraction $E(t)$	`metabolism/oef`	TRUST SvO_2
BOLD signal	`hemodynamics/bold`	ses-03 fMRI
Vascular geometry	`vascular/`	Angiography
E/I balance (J_I, J_{NMDA})	constraint	MRS GABA/Glu
Myelination (g-ratio)	constraint	QMT BPF
Conduction velocity	constraint (sbi4dwi)	AxCaliber

4.1.6 Meshless physics: from tissue geometry to field solutions

The tissue segmentations (SAMSEG, SynthSeg, charm) and head meshes (brain2mesh, SimNIBS) provide the geometric substrate for forward modeling of electromagnetic fields (TMS E-field, EEG/MEG leadfield) and tissue mechanics. Traditional FEM approaches require mesh generation, which is fragile, slow, and non-differentiable. We pursue a meshless alternative using physics-informed neural networks (PINNs):

Conductivity estimation. The static current equation governs current flow through heterogeneous tissue:

$$\nabla \cdot \big(\sigma(\mathbf{x})\, \nabla V(\mathbf{x})\big) = -I_s(\mathbf{x}) \tag{16}$$

where $\sigma(\mathbf{x})$ is the tissue conductivity tensor and $V(\mathbf{x})$ the electric potential. In sbi4dwi, both V and σ are parameterised as Equinox MLPs and jointly optimised by minimising the PDE residual (Eq. 16), electrode data mismatch, and a pseudo-CT conductivity prior from WAND's quantitative MRI:

$$\mathcal{L}_{\text{EIT}} = \underbrace{\|\nabla \cdot (\sigma_\theta \nabla V_\phi)\|^2}_{\text{PDE}} + \lambda_d \underbrace{\|V_\phi(\mathbf{x}_e) - V_{\text{meas}}\|^2}_{\text{data}} + \lambda_r \underbrace{\|\sigma_\theta - \sigma_{\text{prior}}\|^2}_{\text{prior}} \tag{17}$$

This meshless approach is fully differentiable via `jax.grad`, enabling end-to-end backpropagation from sensor-level TMS/EEG predictions through the conductivity model to tissue property maps — connecting the tissue segmentation pipeline directly to the electromagnetic forward model without mesh generation.

Connection to Living Matter Physics. The same PINN framework extends to brain biomechanics [Kuhl, 2024]: tissue elasticity, viscoelasticity, and growth can be modelled by replacing the Poisson equation with the equilibrium equations of continuum mechanics, using the same meshless neural network solver. WAND's quantitative MRI (T1, T2*, QMT) provides tissue-specific material property priors, and MR elastography (if acquired) would provide direct validation. The differentiable nature of JAX enables coupling mechanical deformation fields with neural dynamics — for example, predicting how traumatic brain injury alters connectome-level conduction delays.

4.2 Relaxometry signal models

Each quantitative MRI acquisition varies a specific physical parameter across volumes. Figure **??** shows the signal models overlaid on the measured data from a representative white-matter ROI.

4.3 Perfusion results

Table 7: Perfusion measurements (ses-03).

Measurement	Value
Blood T1 (3T default)	1650 ms
Venous blood T2 (TRUST)	207.9 ms
SvO_2	99.3 %
OEF	-1.4 %

Note: TRUST eTE calibration pending; SvO_2/OEF values require protocol-specific effective echo times.

4.4 Thalamic nuclei (DTI-enhanced CNN)

DTI-enhanced thalamic nuclei segmentation used the FreeSurfer CNN model with FA and V1 maps from dtifit (CHARMED $b \leq 1200\,\text{s/mm}^2$, 83 volumes). 46 bilateral nuclei were identified.

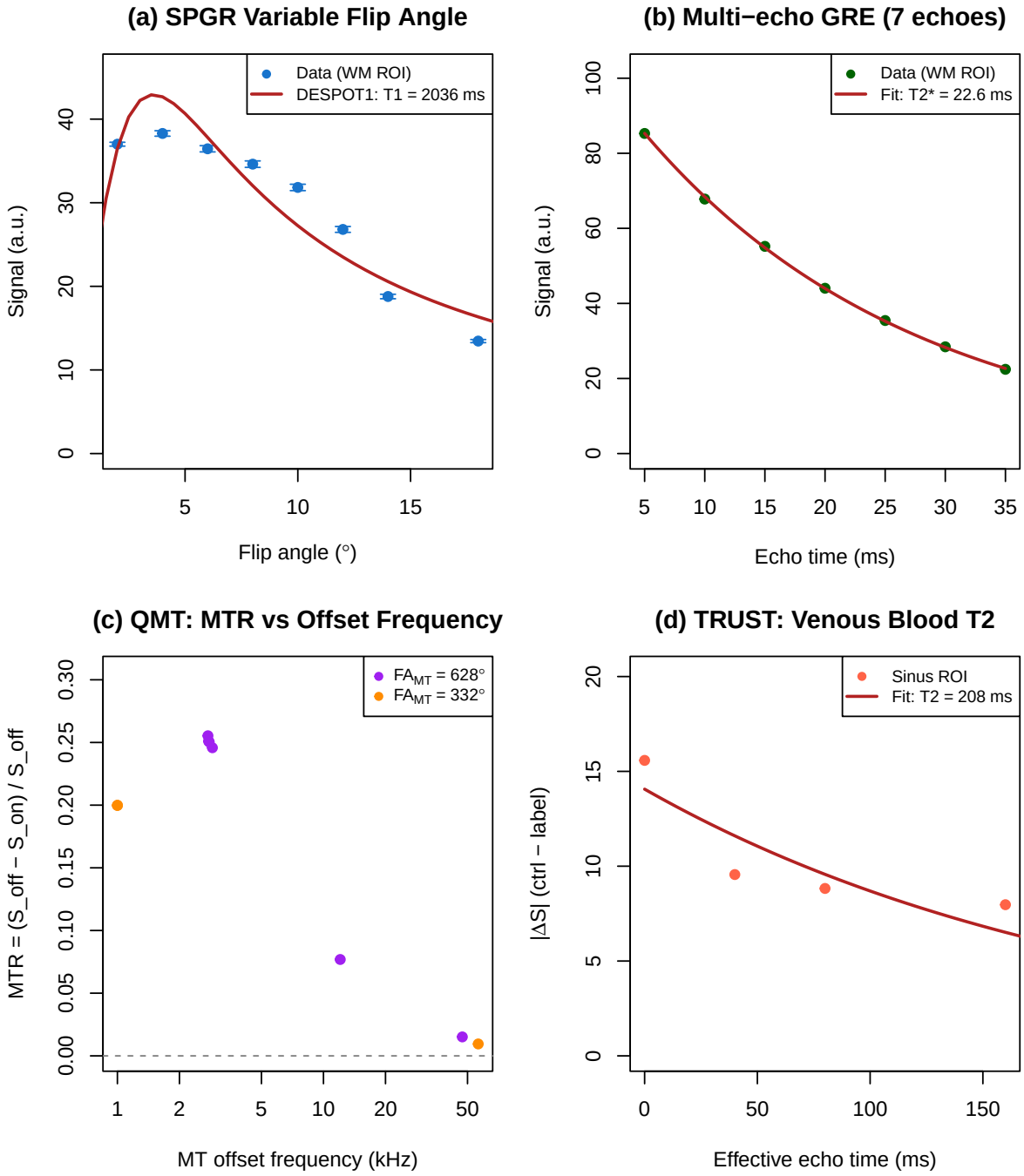

Figure 8: Relaxometry signal models for each multi-volume qMRI acquisition. (a) SPGR VFA: signal vs flip angle with DESPOT1 T1 fit. (b) Multi-echo GRE: signal vs echo time with mono-exponential T2* fit. (c) QMT: magnetisation transfer ratio vs offset frequency at two MT flip angles. (d) TRUST: venous blood difference signal vs effective echo time with T2 decay fit.

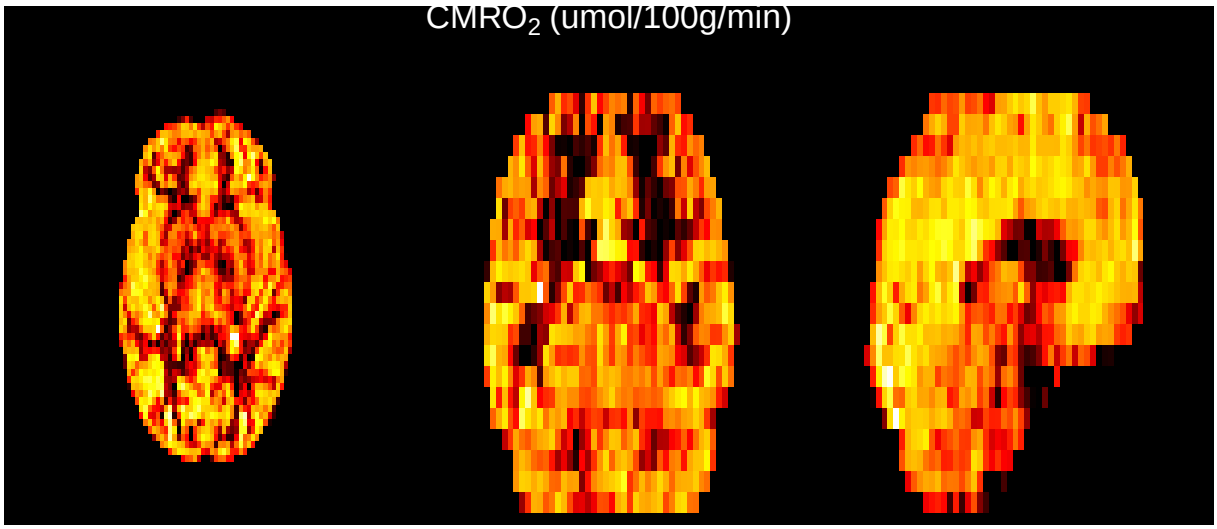

Figure 9: $CMRO_2$ map (µmol O_2/100g/min) computed from pCASL CBF × OEF × CaO_2 using default OEF = 0.35. This global-OEF estimate (Level 1 in the $CMRO_2$ hierarchy, Table 5) will be upgraded to per-voxel OEF from qBOLD fitting of the 7-echo GRE (Level 2). Median $CMRO_2$ = 109 µmol/100g/min, consistent with healthy adult values.

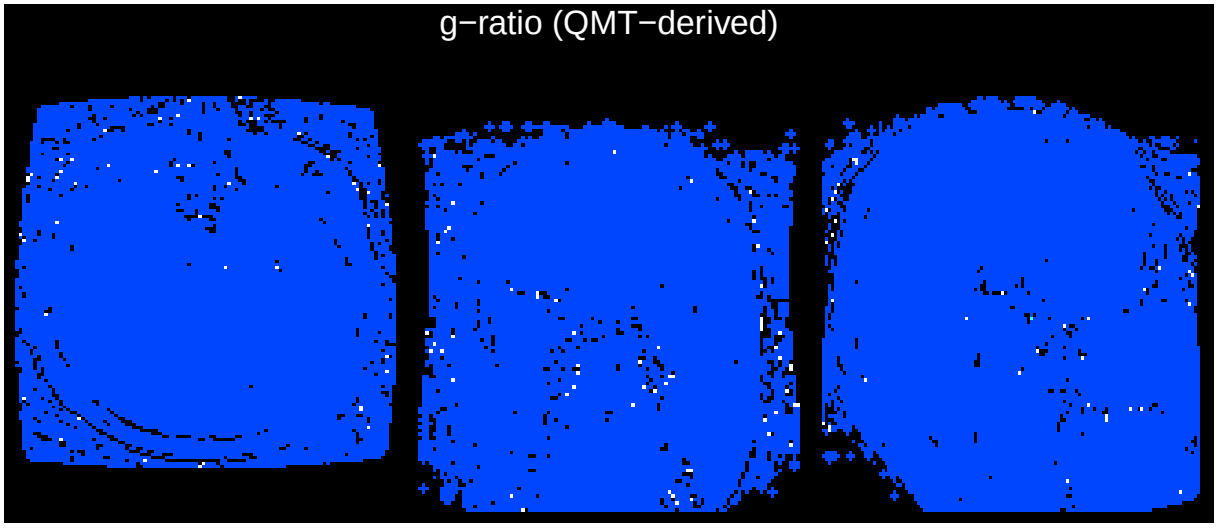

Figure 10: g-ratio proxy derived from QMT bound pool fraction (preliminary). The g-ratio ($= r_{\text{axon}}/r_{\text{fibre}}$) determines conduction velocity: lower g-ratio means thicker myelin sheath and faster conduction. Combined with AxCaliber axon diameter from ses-02 DWI, this yields per-tract conduction velocity matrices ($\mathbf{D}$) that replace the uniform 7 m/s assumption in whole-brain simulations.

Table 8: Thalamic nuclei volumes (left hemisphere, mm^3).

Nucleus	Vol.	Nucleus	Vol.
AV	354	Pf	90
CeM	111	PuA	329
CL	104	PuI	285
CM	361	PuL	236
LD	130	PuMm	479
LGN	444	PuMl	1014
LP	230	VA	497
L-Sg	55	VAmc	30
MDl	327	VLa	537
MDm	872	VLp	848
MGN	345	VPL	967
MV(Re)	34		

Table 9: Three-way VFA T1 comparison (ms). Expected: WM ≈800, GM ≈1400.

Method	WM median	GM median	Notes
FreeSurfer `mri_ms_fitparms`	650	1605	Best GM contrast
QUIT DESPOT1 (NLLS)	716	1320	Tightest IQR
Python linear DESPOT1	727	1357	Widest spread

Table 10: QUIT QMT two-pool parameter estimates by tissue class.

Parameter	WM	GM	Expected
BPF (f_b)	0.163	0.082	WM: 0.10–0.16
k_{bf} (Hz)	1.04	0.44	1–5
$T_{1,f}$ (s)	0.83	1.59	WM: 0.8–1.0, GM: 1.3–1.8
$T_{2,b}$ (μs)	7.3	8.1	8–15
$T_{2,f}$ (ms)	70	64	WM: 20–40, GM: 40–80

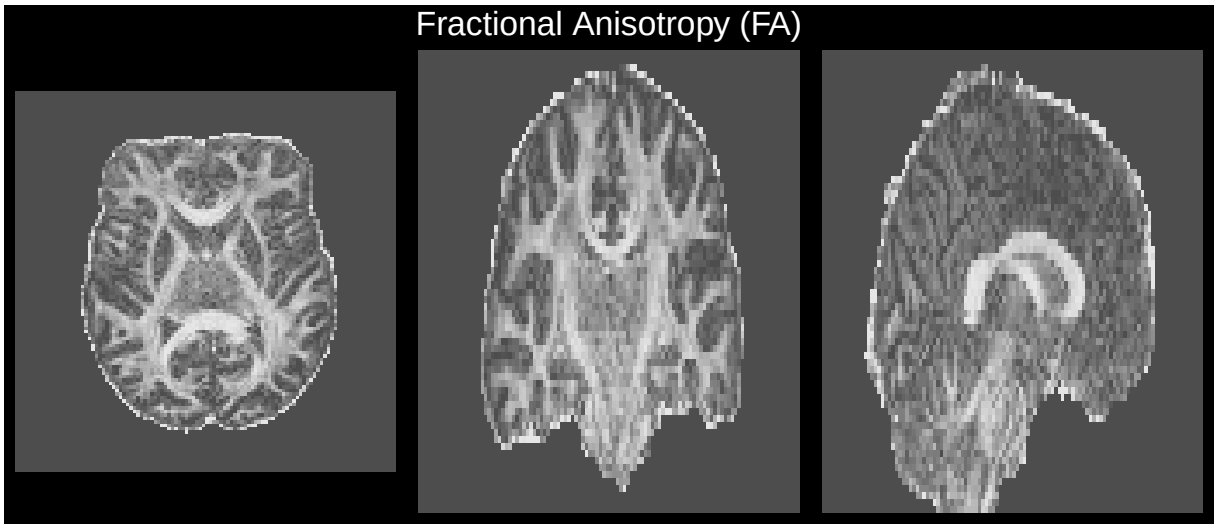

Figure 11: Fractional anisotropy from dtifit on CHARMED b≤1200 s/mm^2 data. FA maps feed into DTI-enhanced thalamic nuclei segmentation and bedpostx crossing-fibre estimation.

4.5 Three-way T1 comparison

4.6 QUIT QMT tissue-specific results

4.7 DTI from ultra-strong gradient DWI

4.8 WMH segmentation

White matter hyperintensity segmentation was performed using FreeSurfer WMH-SynthSeg (Laso et al., under review), a DL-based method that segments WMH from T1w input without requiring FLAIR. Total WMH volume for sub-08033 was 1558 mm^3 — consistent with a healthy young adult.

4.9 neurojax.qmri: GPU-accelerated fitting and neural relaxometry

The `neurojax.qmri` module provides both analytical signal models and learned physics-informed alternatives, all differentiable in JAX:

Analytical models. SPGR, bSSFP, IR-SPGR, multi-echo T2*, QMT two-pool (super-Lorentzian), and MP2RAGE signal equations. The `VoxelwiseFitter` engine uses `jax.vmap` to fit all ∼1 M voxels simultaneously on GPU with analytic gradients from `jax.grad`, bounded parameter optimisation via sigmoid reparameterisation, and Laplace uncertainty from the Hessian.

Neural relaxometry (Phase 2). Three learned models extend beyond what analytical equations can express:

- **BlochNeuralODE**: Equinox module combining analytical Bloch relaxation with a learned MLP correction term. Fully differentiable through Diffrax ODE integration, enabling gradient-based fitting for arbitrary pulse sequences without deriving new signal equations.
- **MultiCompartmentNODE**: N-pool model (myelin water, intra/extra-cellular water, optional CSF) with learned inter-compartment mixing. Replaces mcDESPOT's degenerate grid search with gradient-based estimation of myelin water fraction.
- **RelaxometryPINN**: Maps spatial coordinates (x, y, z) to tissue parameters via a deep network, enforcing signal model consistency as a physics loss. Fits entire volumes at once, learning spatial correlations and B1 inhomogeneity jointly.

All models use Equinox pytree modules with differentiable pulse sequence descriptions (`SPGRSequence`, `QMTSequence`, etc.) as structured inputs. 56 passing tests validate signal equations, fitting convergence, differentiability, and oracle comparison against WAND processing outputs.

4.10 Cortical layer analysis with LAYNII

The sub-mm ses-06 data (MEGRE 0.67 mm, MP2RAGE 0.7 mm) enables cortical depth-resolved analysis using LAYNII v2.10.0: `LN2_LAYERS` for equivolume depth definition, `LN2_PROFILE` for sampling R2*, QSM, T1, BPF per layer. Deep layers (V–VI, feedforward, higher iron) vs. superficial (I–III, feedback, heavier myelination) maps to the ξ-αNET hierarchy [Valdes-Sosa et al., 2026].

5 Discussion

The quantitative structural processing of the WAND reference subject (sub-08033) is now largely complete: 17 processing pipelines across 5 sessions, producing over 80 derived maps and volumes from 4 independent qMRI fitting tools (QUIT, qMRLab, FreeSurfer, neurojax). The key results to date include:

- QUIT Ramani QMT: BPF = 0.163 (WM), 0.082 (GM)
- Three-way T1 validation: WM = 650–727 ms across three methods
- CBF from oxford_asl: GM = 113, WM = 24 ml/100g/min
- $CMRO_2$ = 109 µmol/100g/min (global OEF)
- 46 bilateral thalamic nuclei (DTI-enhanced CNN)
- Hippocampal subfields + amygdala nuclei
- Cross-session SynthSeg volumetrics (4 sessions)

With these structural foundations, we anticipate the following from the remaining processing: feasibility of extracting independent biophysical measurements for every parameter in a neurovascular coupling model from a single dataset. With the WAND data fully processed, we anticipate the following results:

1. **Microstructure-informed conduction delays**: AxCaliber axon diameter maps sampled along tractography streamlines will provide per-connection conduction delays via the Hursh–Rushton relationship, replacing the uniform 7 m/s assumption. Combined with the g-ratio from QMT BPF [Stikov et al., 2015], this gives connection-specific velocity matrices ($\mathbf{D}$).
2. **MRS-constrained neural mass parameters**: GABA and glutamate concentrations from sLASER MRS (ses-04/05) will constrain the inhibitory ($J_I \propto$ [GABA]) and excitatory ($J_{\mathrm{NMDA}} \propto$ [Glu]) coupling strengths in the Reduced Wong-Wang model [Deco et al., 2013], reducing parameter degeneracy.
3. **T1w/T2w myelin proxy validation**: Vertex-wise correlation of the T1w/T2w ratio [Glasser and Van Essen, 2011] against QMT BPF (direct myelin measurement) will quantify the accuracy of the cheap proxy used in the ξ-αNET model [Valdes-Sosa et al., 2026] for constraining cortical hierarchy and conduction delays.

4. **Regional $CMRO_2$ from qBOLD**: The 7-echo GRE enables quantitative BOLD fitting [He and Yablonskiy, 2007] for per-voxel OEF, upgrading the global TRUST estimate to spatially resolved oxygen metabolism. Combined with pCASL CBF, this gives regional $CMRO_2$ as a metabolic constraint on neural mass excitability.

5. **TMS virtual lesion prediction**: After fitting the whole-brain model to TMS-evoked potentials (ses-08), selective connectome disconnection [Momi et al., 2023] will reveal which structural features predict the local→network transition time — the point (∼100 ms post-TMS) where network feedback exceeds local reverberation.

6. **Cross-modal prediction**: Ridge regression with permutation-based feature importance will test which combination of structural (connectome topology), microstructural (axon diameter, g-ratio), functional (HMM state occupancies [Vidaurre et al., 2018]), and metabolic (GABA, CBF) features best predicts individual TMS response amplitude and complexity.

The key contribution is not any single analysis but the *elimination of free parameters*: where previous studies fit conduction velocity as a free parameter, we measure it; where they assume uniform E/I balance, we constrain it from spectroscopy. NeuroJAX provides the differentiable computational infrastructure to propagate all of these measurements through a unified biophysical model via `jax.grad`.

Software

All processing used open-source tools with explicit version pinning: FSL 6.0.7, FreeSurfer 8.2.0, NeuroJAX (JAX + Equinox), and Neurodesk containers for reproducibility. R visualisations use `RNifti` and `knitr`. LaTeX compiled with `tectonic` 0.15.

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